DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-II

COMPILER DESIGN

Class: B. E (CSE)
Semester: VI
Academic Year: 2025-2026

Max Marks: 30
Duration: 1hr 30 min
Date: 06-05-2026

Q.NO	PART A Answer All Questions (5 x 2 = 10 M)	Marks	BTL	CO
1	What is a handle in the context of bottom-up parsing? Provide an example to illustrate your answer.	2	2	2
2	Define a shift-reduce conflict. When and why does it occur in parser construction?	2	2	2
3	Differentiate clearly between synthesized and inherited attributes using a practical grammar-based example.	2	4	3
4	List and describe two widely used code optimization techniques. Support your explanation with appropriate examples.	2	2	4
5	Outline the general steps involved in constructing a parse table for an LR parser.	2	2	2
	PART B **Answer Any Two Questions (2*10 = 20 Marks)**			
6	a) List and explain the key operations performed during bottom-up parsing such as shift, reduce, accept, and error handling.	4	2	2
	a) Develop the SLR (1) parsing table for the provided grammar and describe the process of filling the ACTION and GOTO entries. E -> T + E/ T T -> id	6	3	2
7	a) Explain the different ways to represent three-address code for the following expression: Y := (a - b) + c * (a - b)	6	3	5
	b) Compare and contrast static runtime environments with stack-based runtime environments in terms of memory allocation and management.	4	4	5
8	a) What is Syntax Directed Translation (SDT)? Demonstrate the SDT rules to convert the infix expression A * (B + C) into postfix notation.	5	3	3
	a) Verify the following grammar is LR (0) or Not. S -> (L) / a L -> L , S/ S	5	4	2